

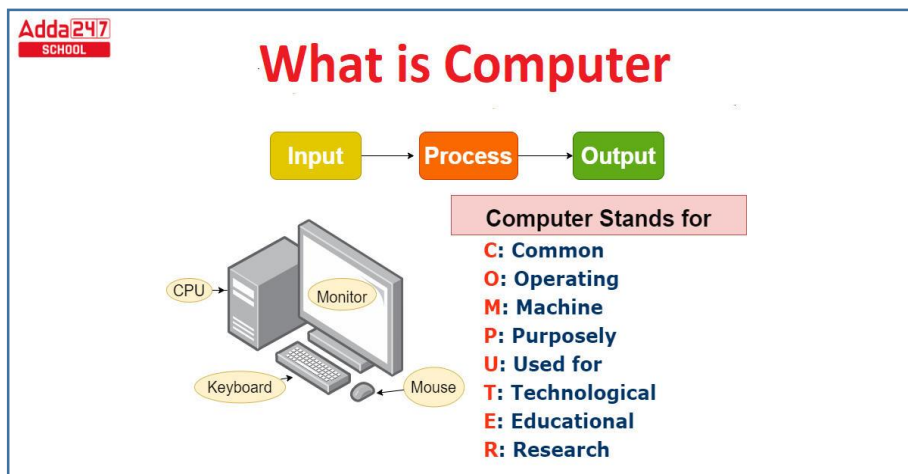
Unit 1: Computer Fundamentals

Computer:-

Computer is an electronic device that processes data according to instructions provided by software programs. It takes input (data), processes it using a central processing unit (CPU), stores information, and produces output (results) to perform various tasks. It is widely used in fields such as education, business, communication, and entertainment for efficient and accurate operations.

A **computer system** is a combination of **hardware and software** that works together to receive data, process it, store it, and produce useful information.

For example, a computer system includes physical components such as the **CPU, keyboard, monitor, and memory**, as well as software such as the **operating system and applications**.



Characteristics and Applications of Computers

1. Characteristics of Computers

The **characteristics of a computer** are the special features that make computers useful and powerful.

1. High Speed

A computer can perform **millions or billions of calculations in a very short time**. It can process large amounts of data much faster than humans.

Example: Calculating the marks and percentage of thousands of students.

2. Accuracy

A computer produces highly accurate results when the **data and instructions given to it are correct**.

Example: Preparing salary sheets and calculating taxes.

3. Diligence

A computer does not become tired or bored. It can perform the **same task repeatedly** with the same efficiency.

Example: Processing thousands of bank transactions continuously.

4. Large Storage Capacity

Computers can store a **large amount of data** and retrieve it whenever required.

Example: Storing documents, photographs, videos, student records, and databases.

5. Automation

Once a computer is given the necessary instructions and data, it can perform tasks **automatically with little or no human intervention**.

Example: Automatic generation of monthly electricity bills.

6. Versatility

A computer can perform **different types of tasks** such as calculations, communication, designing, programming, entertainment, and data processing.

Example: The same computer can be used for teaching, programming, browsing, and watching videos.

7. Reliability

Computers can perform tasks consistently for long periods with a low chance of error caused by fatigue.

Example: Computers are used for continuous monitoring in industries and servers.

8. Multitasking

A computer can handle **multiple tasks at the same time**.

Example: A user can listen to music, browse the internet, download a file, and edit a document simultaneously.

9. Communication

Computers can communicate and exchange information through **computer networks and the Internet**.

Example: Email, video conferencing, online meetings, and instant messaging.

10. Programmability

A computer can be programmed to perform different tasks according to the instructions provided by the user.

Example: Programming a computer to calculate student results or manage inventory.

Applications of Computers

Computers are used in almost every field of modern life.

<i>Field</i>	<i>Applications</i>
<i>Education</i>	✓ Online classes, e-learning, examinations, presentations, digital libraries
<i>Business</i>	✓ Accounting, billing, payroll, inventory, sales management
<i>Banking</i>	✓ ATM, online banking, transaction processing, account management
<i>Healthcare</i>	✓ Patient records, medical imaging, diagnosis support, hospital management
<i>Communication</i>	✓ Email, video calls, social media, messaging
<i>Science & Research</i>	✓ Data analysis, simulations, experiments, scientific calculations
<i>Engineering</i>	✓ CAD, design, simulation, project management
<i>Entertainment</i>	✓ Movies, music, games, video editing
<i>Government</i>	✓ Digital records, taxation, public services, online forms
<i>Transportation</i>	✓ Ticket booking, navigation, traffic control, vehicle management
<i>Industry</i>	✓ Automation, robotics, production control, quality checking
<i>E-commerce</i>	✓ Online shopping, digital payment, order and delivery management
<i>Defense</i>	✓ Communication, simulation, monitoring, information processing
<i>Agriculture</i>	✓ Weather monitoring, crop management, smart irrigation
<i>Home</i>	✓ Online banking, education, shopping, entertainment, smart-home control

Applications of Computers :-

1. **Education:** Computers are used for online classes, e-learning, presentations, and digital libraries.
They help teachers prepare lessons and students learn and research easily.
2. **Business:** Computers are used for accounting, billing, payroll, inventory, and data management.
They help businesses perform daily operations quickly and accurately.
3. **Banking:** Computers are used for ATM services, online banking, account management, and transactions.
They make banking services faster, safer, and available 24/7.
4. **Healthcare:** Computers are used to maintain patient records, medical imaging, diagnosis, and hospital management.
They help doctors and hospitals manage information efficiently.
5. **Communication:** Computers are used for email, video conferencing, messaging, and social media.
They allow people to communicate and share information from anywhere.
6. **Science and Research:** Computers are used for complex calculations, simulations, experiments, and data analysis.
They help researchers process large amounts of scientific data quickly.
7. **Engineering:** Computers are used for designing, modeling, simulation, and technical calculations.
CAD software helps engineers create accurate designs and drawings.
8. **Entertainment:** Computers are used for watching movies, listening to music, playing games, and editing videos.
They provide various forms of digital entertainment.
9. **Government:** Computers are used for maintaining records, taxation, online services, and public administration.
They help government offices provide services quickly and efficiently.
10. **Transportation:** Computers are used for ticket booking, navigation, traffic management, and vehicle control.
They help make transportation systems faster, safer, and more organized.
11. **Industry:** Computers control machines, robots, production processes, and quality checking.
They increase productivity and reduce human effort in manufacturing.
12. **E-commerce:** Computers are used for online shopping, digital payments, order processing, and customer management.
They allow customers to buy products and services through the Internet.
13. **Defense:** Computers are used for communication, monitoring, simulation, and information processing.
They help defense organizations manage and analyze large amounts of information.
14. **Agriculture:** Computers are used for weather forecasting, crop monitoring, irrigation management, and farm planning.
They help farmers improve productivity and make better decisions.
15. **Home:** Computers are used for study, entertainment, online shopping, banking, and communication.
They make many everyday activities easier and more convenient.

Generations and Classification of Computers

Generations of Computers

1. First Generation (1940–1956)

1. First-generation computers used **vacuum tubes** for processing.
2. They used **machine language** for programming.
3. They were very **large, expensive, and slow** compared with modern computers.
4. They consumed a lot of **electricity** and produced excessive heat.
5. Their storage mainly used **magnetic drums**.
6. **Examples:** ENIAC, UNIVAC-I, and EDSAC.

2. Second Generation (1956–1963)

1. Second-generation computers used **transistors** instead of vacuum tubes.
2. They were **smaller, faster, cheaper, and more reliable**.
3. They consumed less electricity and produced less heat.
4. **Assembly language** and high-level languages such as FORTRAN and COBOL were used.
5. **Magnetic core memory** was commonly used.
6. **Examples:** IBM 1401, IBM 7094, and CDC 1604.

3. Third Generation (1964–1971)

1. Third-generation computers used **Integrated Circuits (ICs)**.
2. ICs contained many electronic components on a single chip.
3. Computers became **smaller, faster, more reliable, and less expensive**.
4. Operating systems and **high-level programming languages** became more common.
5. They supported features such as **multiprogramming and time-sharing**.
6. **Examples:** IBM System/360, PDP-8, and PDP-11.

4. Fourth Generation (1971–Present)

1. Fourth-generation computers use **microprocessors** based on VLSI technology.
2. Thousands or millions of components can be placed on a single chip.
3. Computers became **very small, powerful, affordable, and reliable**.
4. Personal computers, laptops, and modern networking developed rapidly.
5. Graphical user interfaces, databases, and advanced software became common.
6. **Examples:** IBM PC, Apple Macintosh, modern desktops, and laptops.

5. Fifth Generation (Present and Beyond)

1. Fifth-generation computers focus on **Artificial Intelligence (AI)** and intelligent computing.
2. They use technologies such as **parallel processing, machine learning, and natural language processing**.
3. They are designed to understand and respond to human-like commands.
4. Robotics, expert systems, speech recognition, and computer vision are important applications.
5. Modern AI systems can learn from data and perform complex tasks.
6. **Examples:** AI assistants, intelligent robots, autonomous systems, and AI-based computers.

Classification of Computers Based on Size

1. Microcomputer

1. Microcomputers are **small computers designed mainly for individual users**.
2. They generally use a **microprocessor** as the CPU.
3. They are relatively inexpensive and easy to use.
4. They are commonly used in homes, schools, offices, and businesses.
5. Examples include **desktop computers, laptops, tablets, and smartphones**.
6. They are also commonly called **personal computers (PCs)**.

2. Minicomputer

1. Minicomputers are **medium-sized computers** that are more powerful than typical microcomputers.
2. They can support **multiple users simultaneously**.
3. They were commonly used by universities, laboratories, and medium-sized organizations.
4. They provide greater processing and storage capacity than traditional personal computers.
5. They were often used for **database and industrial applications**.
6. **Examples:** DEC PDP-8 and PDP-11.

3. Mainframe Computer

1. Mainframe computers are **large and powerful computers** designed to process huge amounts of data.
2. They can support **hundreds or thousands of users** simultaneously.
3. They provide high reliability, security, and availability.
4. They are widely used for banking, insurance, airlines, and government systems.
5. They can handle millions of transactions and large databases efficiently.
6. **Examples:** IBM zSeries and IBM Z mainframes.

4. Supercomputer

1. Supercomputers are the **most powerful computers** in terms of computational performance.
2. They perform extremely large and complex calculations at very high speeds.
3. They use thousands of processors working together through **parallel processing**.
4. They are used for weather forecasting, scientific research, simulations, and space research.
5. They are also used in areas such as nuclear research and advanced AI computation.
6. **Examples:** Frontier, Aurora, and Fugaku.

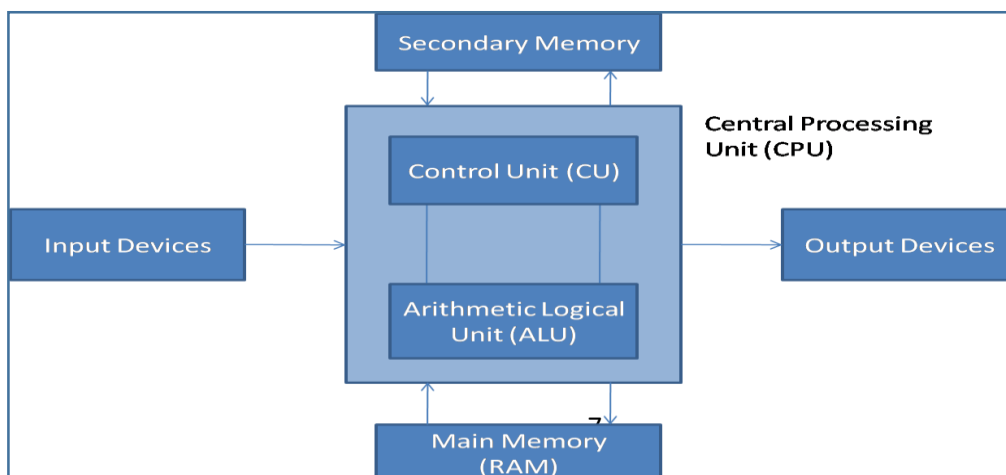
Computer Architecture

Computer architecture refers to the **design, organization, and functional structure of a computer system**. It explains how different components of a computer are connected and how they work together to execute instructions and process data.

Main Components of Computer Architecture

- 1. Input Unit**
It accepts data and instructions from the user and converts them into a form that the computer can understand.
Examples: Keyboard, mouse, scanner.
- 2. Central Processing Unit (CPU)**
The CPU is the **main processing unit** of a computer. It executes instructions and controls other components.
It consists mainly of the **ALU, Control Unit, and Registers**.
- 3. Arithmetic Logic Unit (ALU)**
ALU performs **arithmetic operations** such as addition and subtraction.
It also performs **logical operations** such as AND, OR, NOT, and comparisons.
- 4. Control Unit (CU)**
The Control Unit controls and coordinates all operations of the computer.
It fetches instructions from memory, decodes them, and directs other units to execute them.
- 5. Memory Unit**
Memory stores **data, instructions, and intermediate results** required during processing.
It includes primary memory such as **RAM and ROM** and secondary storage such as HDD and SSD.
- 6. Output Unit**
The output unit converts processed information into a form that users can understand.
Examples: Monitor, printer, and speakers.

Basic Computer Architecture



How It Works

Input → Memory → CPU Processing → Memory → Output

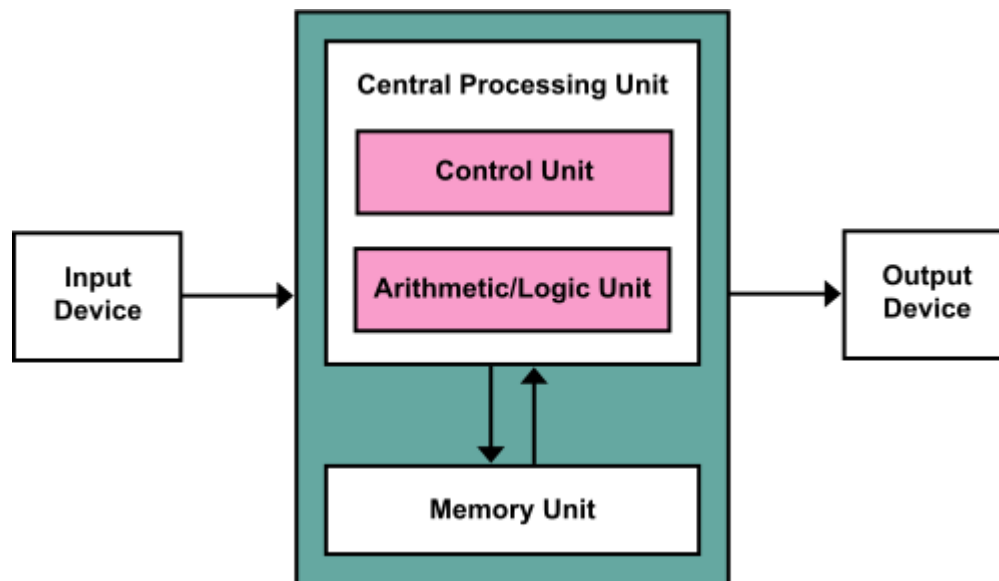
For example, when you calculate **10 + 20**, the computer receives the numbers through input, the CPU/ALU performs the addition, the result **30** is stored temporarily in memory, and the result is displayed through the output device.

Von Neumann Model

The **Von Neumann model** is a basic computer architecture proposed by **John von Neumann**. It describes how a computer stores **data and instructions in the same memory** and processes them using the CPU.

The Von Neumann model is a computer architecture in which data and program instructions are stored in the same memory and processed by the CPU using the fetch, decode, and execute cycle.

Basic Diagram



Main Components

1. Memory Unit

Memory stores **both instructions and data** in the same storage area. The CPU retrieves instructions and data from memory when needed.

2. Control Unit (CU)

The Control Unit controls the activities of the computer. It **fetches instructions, decodes them, and controls their execution.**

3. Arithmetic Logic Unit (ALU)

The ALU performs **arithmetic operations** such as addition and subtraction and **logical operations** such as comparison and Boolean operations.

4. Registers

Registers are **small, high-speed storage locations inside the CPU.** They temporarily hold instructions, data, addresses, and intermediate results.

5. Input Unit

The input unit accepts **data and instructions from the user** and sends them to memory or the CPU for processing.

6. Output Unit

The output unit presents the **processed results** to the user through devices such as monitors and printers.

Working of Von Neumann Model

The model follows the **fetch–decode–execute cycle**:

1. **Fetch:** CPU fetches an instruction from memory.
2. **Decode:** Control Unit interprets the instruction.
3. **Execute:** ALU or another CPU component performs the required operation.
4. **Store:** The result is stored in memory or a register.
5. The CPU repeats the cycle for the next instruction.

Important Features

- **Single memory** stores both data and instructions.
- Uses a **stored-program concept.**
- Instructions are normally executed **sequentially.**
- CPU consists mainly of **ALU, Control Unit, and registers.**
- Communication occurs through **buses.**
- It provides a simple and flexible structure for general-purpose computers.

Main Limitation: Von Neumann Bottleneck

Because **instructions and data share the same memory and communication path**, the CPU may have to wait while transferring them. This limitation is called the **Von Neumann bottleneck.**

Introduction to Number Systems in Computer

A **number system** is a method of representing numbers using a specific set of digits or symbols. In computers, number systems are used to represent, store, and process **data and instructions**.

Computers mainly use the **binary number system** because electronic circuits work with two states: **ON (1)** and **OFF (0)**.

A number system is a systematic way of representing numbers using a specific set of digits and a base. In computers, binary, octal, decimal, and hexadecimal number systems are commonly used.

Types of Number Systems

There are four commonly used number systems in computers:

Number System	Base (Radix)	Digits Used	Example
Binary	2	0, 1	$(1011)_2$
Octal	8	0–7	$(25)_8$
Decimal	10	0–9	$(125)_{10}$
Hexadecimal	16	0–9, A–F	$(2A)_{16}$

1. Binary Number System

The **binary system** has a base of 2 and uses only **0 and 1**.

- Each digit is called a **bit**.
- It is the fundamental number system used by computers.
- **Example:** $(10101)_2$

Example conversion:

$$\begin{aligned}(1011)_2 &= 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\ &= 8 + 0 + 2 + 1 \\ &= 11_{10}\end{aligned}$$

Example:

$(1011)_2$ can be written as

$$(1011)_2 = 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$(1011)_2 = 1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1$$

$$(1011)_2 = 11 \text{ (in decimal)}$$

2. Octal Number System

The **octal system** has a base of **8** and uses digits from **0 to 7**.

- One octal digit represents **3 binary bits**.
- It provides a shorter representation of binary numbers.
- **Example:** $(725)_8$

- **Example:**
 - $(325)_8$ can be written as

$$(325)_8 = 3 \times 8^2 + 2 \times 8^1 + 5 \times 8^0$$

$$(325)_8 = 192 + 16 + 5$$

$$(325)_8 = 213 \text{ (in decimal)}$$
-

3. Decimal Number System

The **decimal system** has a base of **10** and uses digits from **0 to 9**.

- It is the number system commonly used by humans in everyday life.
- Each position has a power of 10.
- **Example:** $(256)_{10}$

Example:

$$256 = 2 \times 10^2 + 5 \times 10^1 + 6 \times 10^0$$

Example:

10285 can be written as

$$10285 = (1 \times 10^4) + (0 \times 10^3) + (2 \times 10^2) + (8 \times 10^1) + (5 \times 10^0)$$

$$10285 = 1 \times 10000 + 0 \times 1000 + 2 \times 100 + 8 \times 10 + 5 \times 1$$

$$10285 = 10000 + 0 + 200 + 80 + 5$$

$$10285 = 10285$$

4. Hexadecimal Number System

The **hexadecimal system** has a base of **16**.

It uses:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

Here, **A = 10, B = 11, C = 12, D = 13, E = 14, F = 15**.

- One hexadecimal digit represents **4 binary bits**.
- It is widely used in programming, memory addresses, machine code, and color codes.
- **Example:** $(2F)_{16}$

- **Example:**
- $(2F)_{16}$ can be written as
$$(2F)_{16} = 2 \times 16^1 + F \times 16^0$$
$$(2F)_{16} = 2 \times 16 + 15 \times 1$$
$$(2F)_{16} = 32 + 15$$
$$(2F)_{16} = 47 \text{ (in decimal)}$$

Positional Value

The value of a digit depends on its **position** and the **base/radix** of the number system.

For a number:

$(1011)_2$

1	0	1	1
↓	↓	↓	↓
2^3	2^2	2^1	2^0

Therefore:

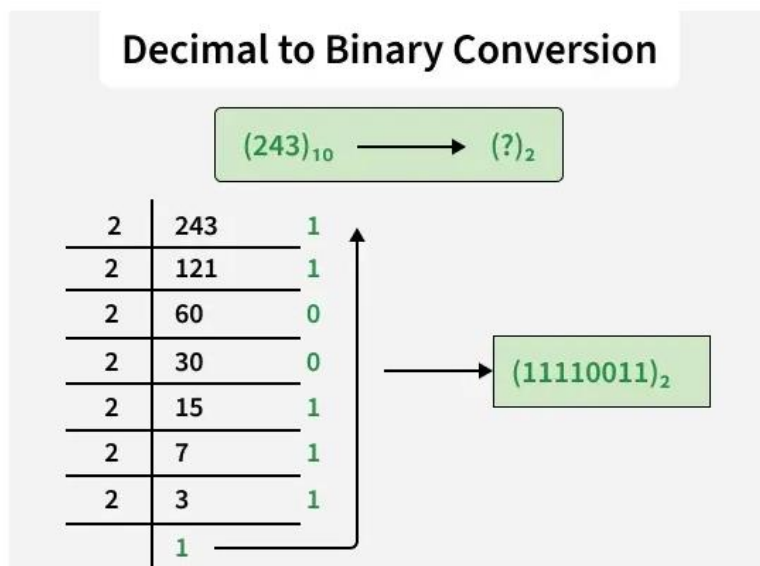
$$1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1 = 11$$

Why Number Systems Are Important

1. They help computers **represent data and instructions**.
2. Binary is used directly by **digital electronic circuits**.
3. Octal and hexadecimal make long binary numbers **easier to read and write**.
4. Decimal is convenient for **human interaction and calculations**.
5. Number-system conversions are important in **computer programming, networking, digital electronics, and computer architecture**.

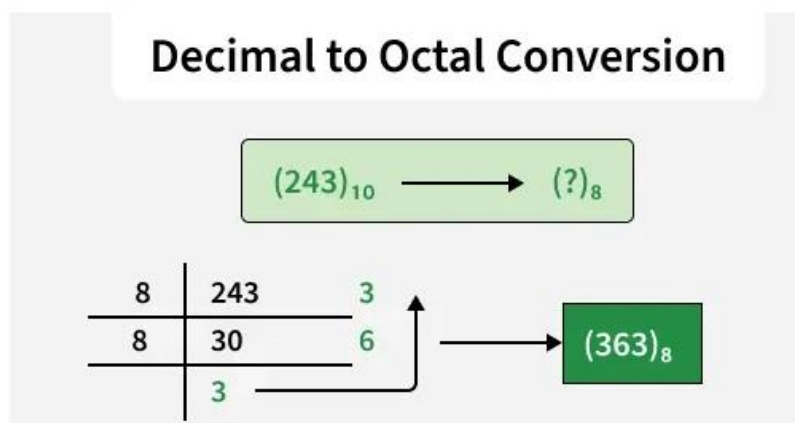
Decimal to Binary Conversion

To see how a decimal number is converted to binary, consider the following example:



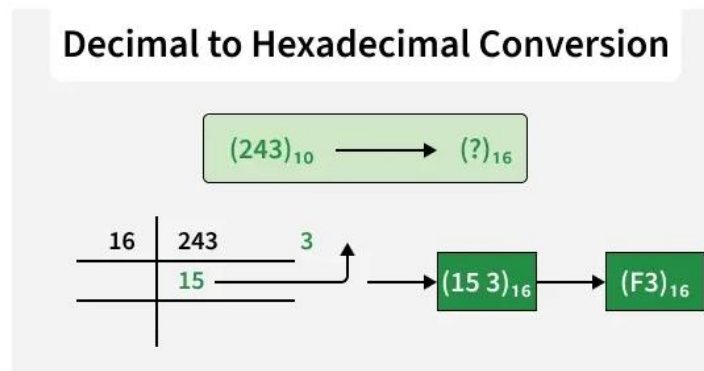
Decimal to Octal Conversion

Let's look at an example of converting a decimal number to octal:



Decimal to Hexadecimal Conversion

Here's an example of converting a decimal number to hexadecimal:



Hexadecimal uses base-16, with digits 0–9 and A–F (where A = 10, B = 11, C = 12, D = 13, E = 14, F = 15).

Example 1. Convert $(1101)_2$ into a decimal number.

Solution: Given a binary number $(1101)_2$.

Now, multiplying each digit from MSB to LSB with reducing the power of the base number 2.

$$1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 8 + 4 + 0 + 1$$

$$= 13$$

Therefore, $(1101)_2 = (13)_{10}$

Octal to Decimal:

To convert octal to decimal, we multiply the digits of octal number with decreasing power of the base number 8, starting from MSB to LSB and then add them all together.

Example 2: Convert 22_8 to decimal number.

Solution: Given, 22_8

$$2 \times 8^1 + 2 \times 8^0$$

$$= 16 + 2$$

$$= 18$$

Therefore, $22_8 = 18_{10}$

Hexadecimal to Decimal:

Example 3: Convert 121_{16} to decimal number.

Solution: $1 \times 16^2 + 2 \times 16^1 + 1 \times 16^0$

$$= 16 \times 16 + 2 \times 16 + 1 \times 1$$

$$= 289$$

Therefore, $121_{16} = 289_{10}$

Example: Convert $(89)_{16}$ into a binary number.

Solution: From the table, we can get the binary value of 8 and 9, hexadecimal base numbers.

$$8 = 1000 \text{ and } 9 = 1001$$

Therefore, $(89)_{16} = (10001001)_2$

Example: Convert $(214)_8$ into a binary number.

Solution: From the table, we know,

$$2 \rightarrow 010$$

$$1 \rightarrow 001$$

$$4 \rightarrow 100$$

Therefore, $(214)_8 = (010001100)_2$

Practice Problems on Number System Conversion

1. Convert 146_{10} into a binary number system
2. Convert $1A7_{16}$ into the decimal number system
3. Convert $(110010)_2$ into octal number system
4. Convert $DA2_{16}$ into the binary number system
5. Convert 4652_8 into the binary number system

Data Representation: ASCII and Unicode

Data representation means the way a computer **stores and represents data** such as characters, numbers, symbols, images, and instructions in digital form. Since computers work with **binary (0 and 1)**, characters are converted into numerical codes so that they can be stored and processed.

1. ASCII

ASCII (American Standard Code for Information Interchange) is a character-encoding standard used to represent letters, numbers, and symbols.

Main Points of ASCII

1. Standard ASCII uses **7 bits** and can represent **128 characters**.
2. It represents English uppercase and lowercase letters, digits, punctuation, and control characters.
3. **A = 65, B = 66, a = 97, and 0 = 48** in decimal.
4. ASCII can be represented in binary, for example **A = 1000001₂**.
5. Extended ASCII commonly uses **8 bits**, allowing up to 256 values, although different extended-ASCII variants are not all the same.
6. ASCII is widely used for basic English text and is also a foundation for many modern character encodings.

Example

<i>Character</i>	<i>ASCII Decimal</i>	<i>Binary</i>
<i>A</i>	65	1000001
<i>B</i>	66	1000010
<i>a</i>	97	1100001
<i>1</i>	49	0110001

2. Unicode

Unicode is a universal character-encoding system designed to represent characters from **many languages and writing systems** around the world.

Main Points of Unicode

1. Unicode can represent characters from languages such as **English, Nepali, Hindi, Chinese, Arabic, Japanese**, and many others.
2. It also supports **mathematical symbols, technical symbols, and emoji**.
3. Every Unicode character is assigned a unique **code point**, written like **U+XXXX**.

4. For example, **A = U+0041** and the Nepali character **अ = U+0915**.
5. Unicode can be encoded using formats such as **UTF-8, UTF-16, and UTF-32**.
6. **UTF-8** is especially common on the Web and is compatible with ASCII for the basic 128 characters.

Examples

Character	Unicode Code Point
A	U+0041
a	U+0061
अ	U+0915
न	U+0928
😊	U+1F60A

ASCII vs Unicode

Feature	ASCII	Unicode
Full form	American Standard Code for Information Interchange	Unicode
Main purpose	Mainly basic English characters	Characters from many writing systems
Standard ASCII size	7 bits	Uses code points; encoded using UTF-8/16/32
Characters	128	Over 1 million possible code points
Languages	Mainly English	Global languages and symbols
Example	A = 65	A = U+0041
Nepali support	✗ No	✓ Yes
Emoji support	✗ No	✓ Yes

Example

When you type:

Hello

The computer does not store the letters directly as human-readable letters. They are represented using character codes and ultimately stored as **binary data**.

BCD in Data Representation

BCD (Binary Coded Decimal) is a method of representing **decimal digits (0–9)** using **4 binary bits**.

Important: BCD is mainly used for **digits 0–9**, not directly for letters **A–Z or a–z**. Letters are represented using codes such as **ASCII** or **Unicode**.

1. BCD Table for Decimal Digits

<i>Decimal</i>	<i>BCD</i>
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

2. Example: Represent 25 in BCD

Each decimal digit is converted separately:

2 → 0010

5 → 0101

Therefore:

25₁₀ = 0010 0101 (BCD)

⚠ BCD is **not** the same as pure binary.

For example, binary of 25 is **11001**, while BCD is **0010 0101**.

3. What about A–Z and a–z?

For **letters**, we normally use **ASCII**, not BCD.

<i>Character</i>	<i>ASCII Decimal</i>	<i>8-bit ASCII</i>
A	65	01000001
B	66	01000010
C	67	01000011
...	...	...
Z	90	01011010
a	97	01100001
b	98	01100010
c	99	01100011
...	...	...
z	122	01111010

Easy way to remember

BCD → Decimal digits (0–9)

ASCII → Characters (A–Z, a–z, 0–9, symbols)

Unicode → Characters from many languages and writing systems

Example:

25 → **BCD:** 0010 0101

A → **ASCII:** 01000001

a → **ASCII:** 01100001